🗓️ Phase 1 (Weeks 1–4): Engineering Upgrade

Goal: Move from Notebook Data Scientist → AI Backend Engineer

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✅ Week 1 – Production Python Foundations

Focus:

- Project structure (src/, services/, routers/)
- OOP design patterns
- Dependency injection
- Logging best practices
- Environment configs (.env, pydantic settings)

Build:

- Structured FastAPI boilerplate
- Proper folder architecture

Deliverable:

Clean, production-ready FastAPI template (you'll reuse forever)

✅ Week 2 – Async + Streaming

Focus:

- Async/await deeply
- Event loop basics
- Streaming responses
- Handling timeouts
- Retry & exponential backoff

Build:

- Streaming LLM API endpoint
- Proper error handling layer

Deliverable:

Streaming chatbot API

✅ Week 3 – Docker + Caching

Focus:

- Advanced Docker (multi-stage builds)
- Redis caching
- Token counting

- Rate limiting

Build:

- LLM API with:
    - Redis cache
    - Token logging
    - Basic cost tracking

Deliverable:

Cost-aware chatbot API

✅ Week 4 – Cloud Deployment

Focus:

- Deploy to AWS/GCP (your choice)
- Environment secrets
- CI/CD basics
- Logging in production

Build:

- Deploy your chatbot API publicly

Deliverable:

Live hosted GenAI service

🗓 Phase 2 (Weeks 5–10): Advanced RAG Systems

Now we level up from “basic RAG” → “enterprise RAG”.

✅ Week 5 – Retrieval Science

Study:

- Chunking strategies (semantic vs fixed)
- Embedding selection
- Similarity metrics

Rebuild your RAG with:

- Better chunking
- Metadata filtering

✅ Week 6 – Hybrid Search

Learn:

- BM25
- Hybrid retrieval (vector + keyword)

Implement:

- Hybrid RAG system

✅ Week 7 – RAG Evaluation

This is where most engineers fail.

Learn:

- Retrieval precision/recall
- Faithfulness metrics
- Answer relevancy

Build:

- Evaluation pipeline
- Compare chunking methods

✅ Week 8 – Guardrails & Hallucination Control

Study:

- Prompt injection attacks
- Output validation
- Structured outputs

Implement:

- JSON response schema
- Validation layer

✅ Week 9–10 – Production RAG System

Build Portfolio Project #2:

Enterprise Document Assistant

Must include:

- API
- Evaluation metrics
- Hybrid retrieval
- Caching
- Guardrails
- Logging

This project alone can land interviews.

🗓️ Phase 3 (Weeks 11–16): LLM Depth + Future-Proofing

Now we strengthen fundamentals.

✅ Week 11 – Transformer Internals

Study deeply:

- Self-attention math
- Q, K, V intuition
- Positional encoding
- Why hallucinations happen

Don’t memorize — understand.

✅ Week 12 – Open-Source LLMs

Work with:

- Meta LLaMA models
- Mistral AI models

Run locally (quantized).

Understand:

- Context window
- Memory usage
- Quantization

✅ Week 13 – Fine-Tuning Basics

Learn:

- LoRA
- PEFT
- When fine-tuning is better than RAG

Fine-tune small model on custom dataset.

✅ Week 14 – Inference Optimization

Study:

- Batching
- Quantization

- Latency optimization

Deploy optimized model endpoint.

✅ Week 15 – AI System Design

Study:

- Multi-agent systems
- Orchestration
- Tool calling
- Structured function calling

✅ Week 16 – Capstone Project

Build Portfolio Project #3:

AI Research Assistant with:

- RAG
- Tool calling
- Structured outputs
- Evaluation metrics
- Cost tracking
- Cloud deployment

This makes you interview-ready.

🧠 Weekly Time Allocation (Working Professional Friendly)

Per week (10–12 hrs):

- 4 hrs learning
- 6 hrs building
- 1 hr reading research/blogs
- 1 hr refactoring code

Building matters more than watching videos.

📈 After 4 Months

You move from:

Data Scientist using LLM APIs

To:

Production-Ready Generative AI Engineer